

Statement of Qualifications

Senior Transportation Planner (Temporary Assignment)

Huiyu Lin

1. Education and Experience in Transportation Policy and Planning

I hold a Ph.D. in Geography from Kent State University and a Master of Arts in Geography from SUNY Binghamton, with academic training in transportation planning, Geographic Information Systems (GIS), spatial analysis, and quantitative modeling. My education provided a strong foundation in applying data-driven analysis to transportation planning and policy development.

As a Sustainability Transportation Analyst with Caltrans' Director's Office of Sustainability, I independently develop analytical methodologies and decision-support frameworks that support statewide transportation planning, sustainability, equity, safety, and Vehicle Miles Traveled (VMT) reduction initiatives.

I currently lead the technical development of a statewide Speed Performance research initiative by creating analytical methodologies to identify excessive speeding on the State Highway System and support future speed abatement planning, aligned with the Caltrans Strategic Plan. I also developed the Corridor Prioritization Tool which provided a standardized framework that supports Caltrans' Corridor Tiering Process by helping Districts prioritize and determine appropriate planning tiers in alignment with CAPTI.

I regularly collaborate with multiple Caltrans divisions and offices to support implementation of the Transportation Equity Index (EQI) across transportation planning processes. I have designed and developed analytical use cases that help implement CalSTA's vision to integrate equity into transportation planning and maintenance prioritization, advancing an equitable transportation system in California.

These experiences have strengthened my ability to translate transportation policy objectives into practical analytical methodologies that support statewide planning and data-driven decision-making.

2. Education and Experience with Data Analysis in the Transportation Field, Including GIS and Statistical Software

I have extensive experience applying GIS, programming, and quantitative analysis to transportation planning using ArcGIS Pro, R, Python, SQL, Tableau, StreetLight, and Replica.

At Caltrans, I independently develop analytical workflows that transform complex datasets into repeatable methodologies and practical planning tools. My work includes

developing GIS workflows, automating analyses in R, creating performance metrics, integrating multiple statewide datasets, and preparing maps, dashboards, and technical reports for planners and decision-makers.

I developed a GIS and R workflow integrating Customer Service Request (CSR) data, the Transportation Equity Index (EQI), and demographic information to conduct statewide spatial and temporal analyses. The results identify patterns in maintenance response and provide data-driven evidence to support equity-informed maintenance prioritization.

As part of the Caltrans Strategic Plan, I am developing analytical methods and performance metrics to identify excessive speeding using NPMRDS traffic performance data. I also conduct before-and-after traffic volume analyses using StreetLight and Replica to evaluate transportation project performance and support rural VMT reduction initiatives.

My technical experience allows me to transform transportation data into decision-support tools that improve planning, investment prioritization, and program implementation across Caltrans.

3. Experience with effective communication skills (email, MS Teams, WebEx, phone, in person)

Effective communication is an essential part of my work because nearly every project requires collaboration across multiple Caltrans divisions and technical disciplines.

I regularly communicate with planners, engineers, subject matter experts, managers, and executive staff through email, Microsoft Teams, WebEx, phone calls, in-person meetings, technical reports, presentations, and documentation. I routinely discuss analytical methodologies, explain technical concepts, incorporate stakeholder feedback, and communicate findings to both technical and non-technical audiences.

My work on EQI implementation requires close collaboration with the Director's Office of Equity and Tribal Affairs to develop use cases, implementation strategies, and technical guidance. I also collaborate with the VMT Team to evaluate transportation project performance and work with multiple offices to develop analytical methodologies supporting statewide transportation planning initiatives.

In addition, I authored the technical documentation for the Corridor Prioritization Tool to ensure future users can consistently understand and apply the analytical framework. Through these experiences, I have become a trusted technical resource within the Transportation Analysis Branch, helping colleagues translate complex analytical results into practical planning applications that support collaborative, data-driven decision-making.